QPCODE: 25900755

MAHATMAGANDHIUNIVERSITY,KOTTAYAM
MGU-UGP(HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

1st SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBBT102 – BIOTECHNOLOGY AND HUMAN WELFARE

(2024 ADMISSION ONWARDS)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt atleast one question from each course outcome to enhance their overall Outcome attainability

Duration:1.5 Hour Max.Marks:50

Part A - Answer the Questions in One Word or Sentence
Answer All Questions- Each Question carries 1 mark

1	What are Biopesticides?		[K] / [CO1]
		Biopesticides are pesticides derived from natural materials like animals, plants, bacteria, and minerals.	
2	What is a virus?		[U] / [CO2]
		A virus is a tiny, infectious particle made of genetic material (DNA or RNA) inside a protein coat that can only replicate by invading a living host cell.	
3	How can GM crops affect biodiversity?		[U] / [CO2]
		GM crops can affect biodiversity through **negative impacts** like harming non-target organisms with toxins, reducing food sources for wildlife, and creating herbicide-resistant "superweeds" that disrupt ecosystems.	
4	What is Anthrax?		[K] / [CO1]
		Anthrax is a serious infectious disease caused by the bacterium Bacillus anthracis, which is found naturally in soil and commonly affects farm animals like cattle, sheep, and goats.	
5	What is a compost? [		U] / [CO2]
		Compost is a nutrient-rich, decomposed organic matter used as a fertilizer and soil amendment in gardening and agriculture. It is created from decomposing plant and food waste, such as kitchen scraps and yard trimmings, through a process involving microorganisms	
6	Define genetically modified crops		[U] / [CO2]
		Compost is a nutrient-rich, decomposed organic matter used as a fertilizer and soil amendment in gardening and agriculture. It is created from decomposing plant and food waste, such as kitchen scraps and yard trimmings, through a process involving microorganisms	
7	What is gene-therapy?		[K] / [CO1]
		Gene therapy is a medical treatment that aims to correct or treat genetic diseases by altering a person's genetic makeup, rather than using drugs or surgery.	
8		Define crop management	[U] / [CO2]
		Crop management is the systematic process of growing and maintaining crops to maximize yield, quality, and sustainability through a series of agricultural practices.	

9		What are bioreactor applications in biotechnology?	[A] / [CO3]
		Bioreactors have diverse applications in biotechnology, including large-scale production of pharmaceuticals like vaccines and antibodies, manufacturing food and beverages through fermentation, and producing biofuels, chemicals, and plastic.	
10		What are Patents?	[K] / [CO1]
		A patent is an exclusive right granted for an invention, protecting it from others being able to make, use, or sell it without permission for a limited time.	
			[10x1 = 10]

Part B - Short Answer Questions
Answer any 4 Questions - Each Question carries 3 marks

11		Why is timing important in harvesting?	[U] / [CO2]
		Timing is important in harvesting because it **maximizes yield and quality**, **minimizes losses from pests, weather, and spoilage**, and **optimizes profitability and sustainability**. Harvesting at the correct stage of maturity ensures the highest nutritional value and market price, while harvesting too early or late can lead to immature or overripe crops, increased pest damage, or waste. **Maximize yield and quality** **Maturity:** Harvesting at peak maturity ensures optimal size, flavor, and nutritional content. **Market value:** Produce harvested at the right time is often of higher quality, which can command better prices in the market. **Prevent losses:** Harvesting too early can result in a higher percentage of immature or unfilled grains, lowering yield. Harvesting too late can lead to losses from pests, shattering, or over-ripening.	
12		Which are the major products of diary biotechnology?	[An] / [CO3]
		Fermented dairy products Yogurt: Made using bacterial cultures to ferment milk, changing its texture and flavor. Cheese: Involves the coagulation of milk proteins to form curds, which are then processed into various types of cheese. Cultured buttermilk: A fermented product that uses a starter culture. Kefir: A fermented milk drink with a sour taste. Traditional Indian products: Dahi (curd), Shrikhand, and Lassi are examples of traditional fermented products. **Processed and preserved products** Butter and butter oil (ghee): Created through churning cream. Condensed and evaporated milk: Milk that has had water removed through concentration. Milk powder: Made by drying milk to create a powder that can be easily stored and transported. Ice cream and frozen desserts: Biotechnology helps in creating new flavors, and cereal-based options. Cream: Includes various types like whipping and heavy cream used for different culinary purposes.	
13		How does white biotechnology impact sustainability?	[K] / [CO1]
		Impacts of white biotechnology on sustainability: Reduces waste and pollution: It aims to create processes with minimal waste and uses microorganisms for bioremediation to break down pollutants, making it a cleaner alternative to traditional manufacturing. Decreases fossil fuel dependency: By using biological feedstocks and producing biofuels, it offers a sustainable alternative to fossil fuels, which helps lower carbon dioxide emissions. Uses renewable resources: White biotechnology leverages living organisms and renewable resources like	

		agricultural or forestry wastes instead of fossil resources, making it less reliant on non-renewable materials. Increases energy efficiency: The biological processes it uses are often more energy-efficient and operate at lower temperatures, reducing the overall energy needed for production. Produces biodegradable products: It enables the creation of products like biodegradable bioplastics and detergents with enzymes, which are more easily broken down and have less environmental impact after their use. Supports a circular economy: By turning waste and by-products into valuable products, white biotechnology supports the principles of a circular economy, where resources are kept in use for as long as possible.	
14	What is biocontrol of weeds?		[U] / [CO2]
		Biocontrol of weeds is the use of living organisms, such as insects, fungi, or other pathogens, to suppress the growth of weeds to manageable levels. Instead of chemical or physical removal, biocontrol relies on a weed's natural enemies, which are either introduced from their native habitat or encouraged through habitat management. Examples include using a specific insect to eat an invasive weed or planting a dense cover crop to outcompete weeds. **Working** **Using natural enemies:** Natural enemies like insects, mites, fungi, and bacteria are used to damage or kill weeds. **Host-specificity:** The selected biocontrol agent is carefully chosen to attack only the target weed, not other desirable plants. **Restoring balance:** By introducing or encouraging these agents, the goal is to restore the natural balance that keeps weed populations in check	
15	What are some applications of biotechnology in industry?		[An] / [CO3]
		<u>Biofuels and energy</u> Biofuels: Microorganisms are engineered to produce biofuels like bioethanol and biodiesel from plant materials, reducing dependence on fossil fuels. Biogas: Microbes are used to produce biogas from organic waste. <u>Chemicals and materials</u> Industrial chemicals: Biotechnology produces chemicals like citric acid and lactic acid through microbial fermentation, often more cost-effectively than synthetic routes. Bioplastics: Biodegradable plastics are made from sources like corn and sugarcane. Specialty materials: Genes from organisms like spiders are used to produce strong materials, such as in the development of "spider silk" for applications like body armor <u>Enzymes and industrial processes</u> Enzymes: Large-scale enzyme production is used in various industries. For example, proteases are used in detergents, and enzymes are used to improve processes in the food and textile industries. Fermentation: This is a core application, used to make products like bread, beer, and wine, as well as some cheeses and other processed foods.	
16	Write down the major periods in history of biotechnology		[K] / [CO1]
		The three major periods in the history of biotechnology are Ancient Biotechnology (pre-1800), characterized by early practices like domestication and fermentation; Classical Biotechnology (1800-1945), marked by scientific breakthroughs such as genetics, vaccines, and antibiotics; and Modern Biotechnology (1945-present), defined by the manipulation of genetic material through technologies like DNA sequencing and genetic engineering. **<u>Ancient Biotechnology (Pre-1800)</u>** This era is defined by the earliest human uses of biological processes without a scientific understanding of them. Examples:	

		Food and agriculture: Domesticating animals, cultivating crops, and the use of fermentation for products like cheese, yogurt, and alcoholic beverages. Medicine: Using moldy bread for treating infections and early forms of selective breeding. **Classical Biotechnology (1800-1945)** This period saw the rise of scientific principles and key discoveries that laid the groundwork for modern biotechnology. Examples: Genetics: Gregor Mendel's laws of inheritance and the discovery of DNA's components. Microbiology: Louis Pasteur's work on fermentation and the development of pasteurization, and Robert Koch's techniques for isolating bacteria. Medicine: The development of the first vaccines by Edward Jenner and the discovery of penicillin by Alexander Fleming. **Modern Biotechnology (1945-Present)** This era is characterized by a deep understanding of molecular biology and the ability to genetically engineer organisms. Key developments: DNA: The discovery of the structure of DNA by Watson and Crick in 1953 was a turning point. Genetic Engineering: The development of recombinant DNA technology in the 1970s allowed genes to be transferred between organisms. Genomics: The Human Genome Project, completed in 2003, has opened the door to new treatments and a deeper understanding of disease. Other advancements: The creation of genetically modified organisms (GMOs) and the development of new therapies like gene therapy and personalized medicine.	
			[4x3 = 12]

Part C - Short Essay Questions
Answer any **3** Questions - Each Question carries **6** marks

17	Elaborate on the methods of antibiotic production		[A] / [CO3]
		The industrial production process ➢ Fermentation: The process begins with a microorganism, often a genetically modified strain, that produces an antibiotic. It is cultivated in a large container (a fermentor) filled with a carefully controlled liquid growth medium that provides all the necessary nutrients. ➢ Optimization: Key parameters like oxygen concentration, temperature, and pH are constantly monitored and adjusted to ensure optimal growth and antibiotic production. This is a batch process that can take several days. ➢ Extraction: Once the fermentation is complete and the antibiotic is produced, it needs to be separated from the microorganisms and the rest of the broth. ➢ Purification: The extracted antibiotic is then purified, often involving processes like filtration, ion exchange, or chemical precipitation, to a crystalline form. ➢ Semi-synthetic production: Some antibiotics are further modified through chemical synthesis after the initial fermentation and purification to enhance their effectiveness or reduce side effects.	
18	Write down the management of E-wastes		[E] / [CO4]

		Effective e-waste management involves reducing, reusing, and recycling, and ensuring proper handling of hazardous materials. **Before disposal** Reduce and reuse: Minimize e-waste by buying only what you need, choosing energy-efficient products, and finding new uses for old devices. Repair and refurbish: Extend the life of your electronics by repairing them or refurbishing them to be used again. Donate: Give functional electronics to charities or sell them to reuse stores. Back up data: Use cloud storage to back up data from old devices, eliminating the need for hardware storage that will become e-waste. **For disposal** Find a responsible recycler: Look for certified e-waste recyclers or collection centers in your area. Check store take-back programs: Some retail stores have programs for recycling old electronics. Use municipal collection: Check with local authorities for specific e-waste collection events or designated drop-off points. Schedule a pickup: Some e-waste management centers offer pickup services. **During the recycling process** Segregation: Carefully separate e-waste into categories (e.g., batteries, circuit boards, appliances) to ensure safe handling. Safe storage: Store e-waste in covered, ventilated, and leak-proof containers, away from flammable materials, until it can be transported. Professional treatment: E-waste is processed through dismantling, pre-processing, and recovery to extract valuable materials and safely dispose of hazardous components, often involving processes like acid baths or advanced material recovery.	
19		How does Gold Biotechnology contribute to modern biological research and medicine?	[K] / [CO1]
		"Gold biotechnology," referring to the application of bioinformatics and nanotechnology to biology, contributes to modern research and medicine by accelerating drug discovery, enabling personalized medicine through genetic analysis, and improving diagnostics using gold nanoparticles. It provides computational tools to analyze vast biological data, design new drugs by predicting molecular interactions, and uses nanomaterials for more sensitive detection of disease markers. **Bioinformatics** ("Gold Biotechnology" - computational aspect) Drug discovery and development: Bioinformatics uses computer simulations to screen millions of compounds and predict their interactions with biological targets, drastically speeding up the drug discovery process. It also aids in designing primers for PCR and understanding protein-drug interactions. Personalized medicine: By analyzing an individual's complete genetic information, it allows for predictions of disease risks and the development of personalized preventive measures and treatments. Genomic research: It helps in the large-scale analysis of genomes, the generation and simulation of new functional genes, and tracing the evolutionary history of genes by comparing DNA sequences. **Nanotechnology ("Gold" - nanoparticles)** Advanced diagnostics: Gold nanoparticles are used in techniques like immune electron spectroscopy for highly sensitive and specific identification of pathogens and disease-related antigens. Therapeutic potential: Nanomaterials can be used to create highly specific electronic sensors for detecting biomolecules associated with diseases. Research tools: The high electron density of gold particles makes them useful for visualizing biological interactions using methods like high-resolution transmission electron microscopy (HRTEM).	
20		What is activated sludge, and how does it work in wastewater treatment?	[E] / [CO4]
		Activated sludge is a biological treatment process that uses a concentrated suspension of microorganisms (bacteria, fungi, and protozoa) to break down organic matter and nutrients in wastewater. It works by aerating the wastewater to provide oxygen for the microbes, which then consume the pollutants, creating a clean	

		effluent. The process involves an aeration tank, a secondary clarifier to separate the solids from the liquid, and the recycling of a portion of the settled sludge back into the aeration tank to maintain the active microbial population. **How it works in wastewater treatment** Aeration tank: Wastewater is mixed with the activated sludge in a large tank. Air or oxygen is continuously supplied, creating an aerobic environment that stimulates the microorganisms. Microbial action: The microorganisms consume the organic pollutants in the wastewater, breaking them down into carbon dioxide, water, and new cell material. This process converts soluble waste into settleable flocs (clumps of microbes). Secondary clarifier: The mixture from the aeration tank flows into a secondary clarifier. Here, the activated sludge flocs settle to the bottom due to gravity, separating them from the now-treated water. Sludge recycling: A portion of the settled sludge is returned to the aeration tank to replenish the microbial population and continue the process. Excess sludge removal: The remaining excess sludge is removed for further treatment and disposal. Final effluent: The treated water that overflows from the secondary clarifier is the clean effluent, which can be further disinfected and released.	
21		What are the challenges in addressing the health-environment nexus, and how can these challenges be overcome?	[U] / [CO2]
		Challenges in addressing the health-environment nexus include inadequate monitoring and data, a lack of integrated policy and systemic change, difficulty translating research into practice, insufficient resources and funding, and the disproportionate impact on vulnerable populations. Other challenges include overcoming the interdisciplinary gap between scientific fields and political decision-making, and addressing specific problems like pollution, climate change, and biodiversity loss, which require systemic shifts in resource management and behavior. **Overcoming the challenges** Enhance multisectoral collaboration: Strengthen health systems: Promote prevention and awareness: Implement nature-based solutions: Strengthen research and data:	
			[3x6 = 18]

Part D - Essay Type Questions Answer any **1** Question Question carries **10** marks			
22		How has modern biotechnology revolutionized agriculture compared to conventional methods?	[K] / [CO1]
		Modern biotechnology has revolutionized agriculture by enabling the creation of crops with enhanced traits like pest resistance, improved nutritional value, and tolerance to environmental stresses, leading to increased yields and more sustainable farming practices. Key advancements include genetic engineering to develop traits such as insect resistance (e.g., Bt cotton) and nutrient fortification (e.g., Golden Rice), as well as the use of molecular markers, CRISPR technology, and microbial biotechnology to accelerate breeding and improve soil health.	
		Key revolutions in agriculture ❖ Enhanced crop traits: Biotechnology allows for the precise modification of crops to be resistant to pests and diseases, reducing the need for chemical pesticides. Example: Crops like Bt cotton produce their own insecticide, leading to higher yields and reduced chemical use. ❖ Improved nutritional value: Crops can be genetically engineered to have higher levels of essential vitamins and minerals to combat malnutrition. Example: Golden Rice is engineered to produce beta-carotene, a precursor to vitamin A, to address deficiency in certain populations.	

		❖ Increased stress tolerance: Biotechnology is used to develop crops that can withstand harsh environmental conditions, such as drought, salinity, or extreme temperatures, which is crucial for climate resilience. ❖ Higher and more efficient yields: By incorporating desirable traits like faster growth and higher yield potential, biotechnology helps increase the amount of food that can be produced per acre. ❖ More sustainable practices: ❖ Reduced chemical use: The development of pest-resistant crops and the use of bio-pesticides and bio-herbicides decreases reliance on synthetic chemicals, which is better for the environment. ❖ Improved soil health: Microbial biotechnology uses beneficial microorganisms to improve soil fertility and nutrient uptake. ❖ Faster and more precise breeding: Biotechnology tools like molecular markers and gene editing (e.g., CRISPR) allow scientists to identify and transfer beneficial genes much more quickly and precisely than traditional breeding methods.	
23	Explain the principle and application of biofilters in waste water treatment		[E] / [CO4]
		Biofilters use a biofilm of microorganisms on a filter media to degrade pollutants in wastewater. The principle involves physical separation, adsorption, and biodegradation as wastewater flows through the media, where bacteria consume organic matter. Applications include treating domestic and industrial wastewater, as well as pre-treatment for drinking water, with common types including trickling filters, intermittent sand filters, and constructed wetlands.	
		<u>Principles</u> ➢ Biofilm formation: Microorganisms attach to the surface of the filter media (like sand, gravel, or plastic) and form a living layer called a biofilm. ➢ Pollutant degradation: As wastewater flows through the biofilter, pollutants are removed through a combination of physical filtration, adsorption onto the media, and biological degradation by the microorganisms in the biofilm. ➢ Biological activity: The microorganisms use the organic matter in the wastewater as a food source, consuming and converting pollutants into less harmful substances. ➢ Process parameters: The effectiveness of the biofilter is influenced by factors such as flow rate, organic loading, temperature, pH, and oxygen availability. ➢ Maintenance: Regular cleaning or backwashing is necessary to remove excess solids that can clog the filter and to maintain the activity of the biofilm. **<u>Applications</u>** ➢ Tertiary wastewater treatment: Biofilters can be used after primary and secondary treatment to remove remaining organic matter, which is especially useful in tertiary treatment processes. ➢ Pre-treatment for drinking water: They can be used to remove dissolved organics and suspended solids before water is sent for further purification, which helps prevent the growth of biofilms in distribution systems. ➢ Constructed wetlands: These are a type of biofilter that uses natural or artificial wetland systems to treat wastewater. ➢ Specific pollutant removal: Certain biofilters are designed to target specific pollutants, such as removing volatile organic compounds (VOCs) and heavy metals from wastewater. ➢ Decentralized systems: They are used in smaller-scale applications, such as on-site treatment for individual homes or small communities.	
			(10 × 1 = 10)